

# Standard form, and fractions as decimals

The national plan's "practical and interdisciplinary" lessons, carrying two Cambridge Stage 9 topics the programme does not otherwise reach.

LESSON 34–35

2 × 40 MIN

ALGEBRA 8, CHAPTER I PRACTICAL

STAGE 9 · 1.2, 8.1 [CAMBRIDGE INSERT]

## LESSON CLOCK



Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

## LEARNING OBJECTIVES

- Write large and small numbers in standard form  $a \times 10^n$ ,  $1 \leq a < 10$ .
- Multiply and divide numbers written in standard form.
- Convert a fraction to a terminating or recurring decimal, and back.
- Tell a rational number from an irrational one by its decimal.

## TEXTBOOK

National pp. 62–67

Cambridge Learner's Book  
pp. 14–16, 162–166

Workbook Workbook 1.2, 8.1

## 01 Explanation

### Standard form

## DEFINITION

A number is in **standard form** when it is written as  $a \times 10^n$  with  $1 \leq a < 10$  and  $n$  a whole number.

- $4\,500\,000 = 4.5 \times 10^6$  — the point moves 6 places left, so  $n$  is positive.
- $0.00032 = 3.2 \times 10^{-4}$  — the point moves 4 places right, so  $n$  is negative.

To calculate, handle the numbers and the powers of ten separately:

$$(3 \times 10^5) \cdot (2 \times 10^3) = 6 \times 10^8$$

$$(6 \times 10^5) : (3 \times 10^2) = 2 \times 10^3$$

## TIDY THE ANSWER

$40 \times 10^4$  is not in standard form. Rewrite it as  $4 \times 10^5$ .

## Fractions as decimals

Divide the numerator by the denominator. Exactly one of two things happens:

- the division **stops** — a **terminating** decimal, e.g.  $\frac{3}{8} = 0.375$ ;
- a block of digits **repeats for ever** — a **recurring** decimal, e.g.  $\frac{1}{3} = 0.333... = 0.$

(3) and  $\frac{5}{11} = 0.454545... = 0.(45)$ .

## WHICH ONE?

Cancel the fraction first. If the denominator's only prime factors are 2 and 5, the decimal terminates; otherwise it recurs.  $\frac{7}{40}$  terminates ( $40 = 2^3 \cdot 5$ );  $\frac{7}{30}$  recurs ( $30 = 2 \cdot 3 \cdot 5$ ).

Every fraction gives a terminating or recurring decimal, and every terminating or recurring decimal comes from a fraction. A decimal that never stops *and* never repeats belongs to an **irrational** number —  $\sqrt{2} = 1.41421356..., \pi = 3.14159...$

## Turning a recurring decimal back into a fraction

Multiply by a power of ten so that one whole period shifts past the point, then subtract.

$$x = 0.(7) \Rightarrow 10x = 7.(7) \Rightarrow 9x = 7 \Rightarrow x = \frac{7}{9}$$

For a two-digit period multiply by 100:  $x = 0.(45) \Rightarrow 100x - x = 45 \Rightarrow x = \frac{45}{99} =$

$$\frac{5}{11}$$

## 02 Worked examples

**EXAMPLE 1** Write 0.00047 and 92 000 in standard form.

①  $0.00047 \rightarrow 4.7$

The point moves 4 places right.

②  $= 4.7 \times 10^{-4}$

Moving right gives a negative index.

③  $92\,000 \rightarrow 9.2$

The point moves 4 places left.

④  $= 9.2 \times 10^4$

**ANSWER**

$4.7 \times 10^{-4}$  and  $9.2 \times 10^4$

**EXAMPLE 2** Work out  $(8 \times 10^6) : (4 \times 10^2)$

①  $8 : 4 = 2$

The numbers.

②  $10^6 : 10^2 = 10^4$

Subtract the indices.

③  $= 2 \times 10^4$

Already in standard form.

**ANSWER**

$2 \times 10^4 = 20\,000$

**EXAMPLE 3** Write  $0.\overline{6}$  as a fraction

①  $x = 0.666\ldots$

Name the number.

②  $10x = 6.666\ldots$

One period is one digit, so multiply by 10.

③  $10x - x = 6$

The tails cancel exactly.

④  $9x = 6, x = \frac{6}{9} = \frac{2}{3}$

Cancel.

ANSWER

$$\frac{2}{3}$$

**03 Interactive model**

Ask for an estimate in standard form before any calculation — it catches most place-value slips.

## Standard form and recurring decimals

$0.00058$

- ① Move the point until one non-zero digit is in front:

$5.8$  .

That  
is  
the  
value  
of a.

- ② The point moved

$4$  .  
**places**  
**right**

Right  
 $\Rightarrow$   
negative  
index.

- ③  $5.8 \times 10^{-4}$

Check:  
 $5.8$   
 $\div 10$   
 $000$   
 $=$   
 $0.00058$   
 $\checkmark$

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH             | O'ZBEKCHA           | РУССКИЙ                        |
|---------------------|---------------------|--------------------------------|
| Standard form       | Standart ko'rinish  | Стандартный вид                |
| Power of ten        | O'nning darajasi    | Степень десяти                 |
| Significant figures | Ahamiyatli raqamlar | Значащие цифры                 |
| Terminating decimal | Chekli o'nli kasr   | Конечная десятичная дробь      |
| Recurring decimal   | Davriy o'nli kasr   | Периодическая десятичная дробь |
| Period              | Davr                | Период                         |
| Rational number     | Ratsional son       | Рациональное число             |
| Irrational number   | Irratsional son     | Иррациональное число           |
| Convert             | Aylantirish         | Преобразовать                  |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$0.00062$  in standard form is:

$$6.2 \times 10^{-4}$$

$$6.2 \times 10^4$$

$$62 \times 10^{-5}$$

$$0.62 \times 10^{-3}$$

$(2 \times 10^3) \cdot (5 \times 10^4)$  equals:

$$10 \times 10^7$$

$$1 \times 10^8$$

$$7 \times 10^7$$

$$10^{12}$$

Which fraction gives a recurring decimal?

$$\frac{3}{8}$$

$$\frac{7}{20}$$

$$\frac{5}{6}$$

$$\frac{9}{25}$$

0.(3) as a fraction is:

$$\frac{3}{10}$$

$$\frac{1}{3}$$

$$\frac{3}{99}$$

$$\frac{1}{30}$$

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS



1. Write 5 000 in standard form

ANSWER  $5 \times 10^3$

2. Write 0.007 in standard form

ANSWER  $7 \times 10^{-3}$

3. Write  $3.2 \times 10^4$  as an ordinary number

ANSWER 32 000

4. Write  $6 \times 10^{-2}$  as an ordinary number

ANSWER 0.06

5. Write  $\frac{1}{4}$  as a decimal

ANSWER 0.25

6. Write  $\frac{1}{3}$  as a decimal

ANSWER 0.(3)

7. Write 0.5 as a fraction in lowest terms

ANSWER  $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. Write  $0.00047$  in standard form

ANSWER  $4.7 \times 10^{-4}$

2. Write  $92\,000$  in standard form

ANSWER  $9.2 \times 10^4$

3. Work out  $(3 \times 10^5) \cdot (2 \times 10^3)$

ANSWER  $6 \times 10^8$

4. Work out  $(8 \times 10^6) : (4 \times 10^2)$

ANSWER  $2 \times 10^4$

5. Does  $\frac{7}{40}$  terminate? Give the decimal.

ANSWER Yes —  $0.175$

6. Write  $\frac{5}{11}$  as a decimal

ANSWER  $0.(45)$

7. Write  $0.(6)$  as a fraction

ANSWER  $\frac{2}{3}$

### Hard · stretch and reason

7 PROBLEMS

1. Work out  $(4 \times 10^5) \cdot (5 \times 10^4)$  in standard form

ANSWER  $2 \times 10^{10}$

2. Work out  $(1.2 \times 10^7) : (4 \times 10^3)$

ANSWER  $3 \times 10^3$

3. Write 0.(45) as a fraction in lowest terms

ANSWER  $\frac{5}{11}$

4. Write 0.1(6) as a fraction

ANSWER  $\frac{1}{6}$  — multiply by 10 and by 100, then subtract.

5. Without dividing, say whether  $\frac{9}{48}$  terminates

ANSWER Cancel to  $\frac{3}{16}$ ;  $16 = 2^4$ , so yes.

6. The mass of a grain of sand is about  $6.6 \times 10^{-5}$  kg. Find the mass of  $2 \times 10^6$  grains.

ANSWER  $1.32 \times 10^2 = 132$  kg

7. Is 0.101001000100001... rational? Explain.

ANSWER No — it never stops and never repeats, so it is irrational.

## 07 Homework

### Homework — 6 problems

Algebra 8, pp. 62–67 (practical problems) and Cambridge Workbook 1.2.

- Write 0.000091 and 340 000 in standard form
- Work out  $(5 \times 10^4) \cdot (6 \times 10^3)$  in standard form

3. Work out  $(9 \times 10^8) : (3 \times 10^5)$
4. Write  $\frac{7}{8}$  and  $\frac{4}{9}$  as decimals
5. Write  $0.(8)$  as a fraction
6. Without dividing, decide whether  $\frac{11}{50}$  terminates